Structural Barriers to Scientific Integration: A Metascientific Analysis of Behavioral Economics Fragmentation

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Abstract

Why do 40+ behavioral models exist in isolation despite universal recognition that integration would benefit science and practice? This paper argues that the barrier is structural, not intellectual. Drawing on metascience, sociology of knowledge, philosophy of science, and science of science literatures, we identify five structural barriers: (1) academic incentive structures favoring novelty over synthesis, (2) disciplinary silos with incompatible formalisms, (3) systematic peer review bias against interdisciplinary work, (4) funding structures rewarding narrow hypotheses, and (5) territorial defense of paradigmatic authority. We document evidence from citation network analysis, publication patterns, experimental studies of peer review, and historical case studies.

A central contribution is the *framework versus model distinction*: classical economics (Samuelson 1947) was conceived as a framework but treated as a model, leading to fragmentation when anomalies accumulated. We trace this epistemological confusion through the Chicago School's methodological development and show why it was technologically defensible until recent computational advances made non-separable, context-dependent modeling feasible.

We conclude that integration requires institutional innovation outside traditional academic structures—a prediction consistent with the emergence of integrative frameworks in consulting and policy contexts rather than universities. The Evidence-Based Framework for Economic and Social Behavior (EBF) exemplifies this pattern, built outside academia with different incentive structures and validated through practice rather than publication.

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1 Introduction: The Integration Puzzle

1.1 The Puzzle

Behavioral science has accumulated remarkable knowledge about human decision-making over the past half-century:

- **Psychology:** Transtheoretical Model, Health Action Process Approach, Rubicon Model, Self-Determination Theory, Protection Motivation Theory (6+ stage models of behavior change)
- **Economics:** Becker-Murphy rational addiction, Fudenberg-Levine dual-self, HANK heterogeneous agents, Gabaix inattention models (10+ formal models)
- **Sociology:** Granovetter threshold models, information cascades, SIR contagion dynamics
- **Management:** Lewin change models, Schein organizational culture, PARC, COM-B frameworks
- **Neuroscience:** Dual-process theories, neural autopilot, habit formation mechanisms

Yet these models remain **fragmented**:

- Different disciplines use different terminology for identical concepts
- Mathematical formalisms are incompatible across fields
- No cumulative validation exists across frameworks
- Practitioners receive contradictory advice from different experts

1.2 The Core Paradox

> *Everyone agrees integration is needed. No one is rewarded for doing it.*

This fragmentation persists despite widespread agreement that integration would be valuable. The puzzle is not whether integration is desirable—it clearly is—but why it has not occurred.

1.3 The Answer: Structural Barriers

This paper argues that the lack of integration is **not** due to:

- Insufficient knowledge (we have enough models)
- Technical impossibility (the existence of integrative frameworks proves it's possible)
- Lack of demand (practitioners desperately need unified guidance)

Rather, it is due to **structural barriers** in the academic system that systematically discourage integration while rewarding fragmentation.

1.4 The Hypothesis

We propose that the barrier to integration is *structural*, not intellectual:

> *Academic institutions systematically reward fragmentation and penalize integration through incentive structures, disciplinary boundaries, peer review processes, funding mechanisms, and territorial dynamics.*

This hypothesis has a testable implication: if the barrier is structural rather than intellectual, integration should emerge from contexts with different incentive structures—consulting firms, policy organizations, or research institutes outside traditional academia.

1.5 Contribution

This paper synthesizes evidence from five literatures:

1. Philosophy of science (Kuhn, Popper, Lakatos)
2. Sociology of scientific knowledge (Merton, Bloor, Latour)
3. Economics of science (Stephan, Smaldino & McElreath)
4. Metascience (Ioannidis, Nosek, Open Science Collaboration)
5. Science of science (Fortunato et al., Uzzi et al., Wang et al.)

We document five structural barriers, present quantitative evidence for each, and trace the historical epistemological confusion that contributed to fragmentation.

2 Classical Foundations: Kuhn, Popper, and Lakatos

2.1 Kuhn: Paradigms and Normal Science (1962)

Thomas Kuhn's *The Structure of Scientific Revolutions* provides the foundational framework for understanding why science fragments.

Key Concepts:

- **Paradigm:** A shared framework of concepts, methods, and exemplars
- **Normal Science:** Puzzle-solving within the paradigm
- **Incommensurability:** Different paradigms cannot be directly compared

Implication for Integration:

> "Normal science... is predicated on the assumption that the scientific community knows what the world is like. Much of the success of the enterprise derives from the community's willingness to defend that assumption." (Kuhn, 1962, p. 5)

Kuhn explains why *within*-paradigm work is rewarded: it solves recognized puzzles. *Between*-paradigm work is seen as threatening, not contributing.

2.2 Popper: Falsification and Demarcation (1959)

Karl Popper's falsificationism emphasizes testable hypotheses:

- Science progresses through conjecture and refutation
- Good theories make bold, falsifiable predictions
- Unfalsifiable theories are "not even wrong"

Problem for Integration: Meta-frameworks are harder to falsify than narrow hypotheses. A claim like "this framework unifies 40 models" is harder to test than "Effect X = 0.3 in population P."

The Popperian criterion inadvertently favors fragmentation by rewarding narrow, testable claims over broad integrative ones.

2.3 Lakatos: Research Programmes (1970)

Imre Lakatos attempted to reconcile Kuhn and Popper:

- **Hard Core:** Unfalsifiable central assumptions
- **Protective Belt:** Auxiliary hypotheses that can be modified
- **Progressive vs. Degenerating:** Programmes that predict novel facts vs. those that don't

Application to Integration: An integrative framework can be seen as a new research programme with:

- Hard Core: Central organizing concepts (e.g., complementarity, context-dependence)
- Protective Belt: Specific parameter estimates for each context
- Progressive: Predicts novel facts that component models cannot predict alone

3 Sociology of Scientific Knowledge

3.1 Merton: The Normative Structure of Science (1942)

Robert Merton identified four norms of science (CUDOS):

1. **Communism:** Knowledge is shared
2. **Universalism:** Claims evaluated on merit, not source
3. **Disinterestedness:** Scientists pursue truth, not self-interest
4. **Organized Skepticism:** Claims are scrutinized

Reality Check: Modern science violates all four:

- Communism $\rightarrow$ Proprietary data, paywalled journals
- Universalism $\rightarrow$ Prestige bias, Matthew Effect
- Disinterestedness $\rightarrow$ Career incentives dominate
- Organized Skepticism $\rightarrow$ Peer review is tribal

3.2 The Strong Programme: Bloor (1976)

David Bloor's "strong programme" in sociology of science argues:

- Scientific knowledge is socially constructed
- The same sociological methods explain "true" and "false" beliefs
- Science is not exempt from sociological analysis

Implication: The fragmentation of behavioral science is not a failure of logic but a *social* outcome of institutional structures.

3.3 Latour and Woolgar: Laboratory Life (1979)

Bruno Latour and Steve Woolgar's ethnography of a biology lab showed:

- "Facts" are constructed through social processes
- Inscription devices (instruments, papers) create "black boxes"
- Once a claim becomes a "fact," its social origins are erased

For Integration: Each behavioral model is a "black box" whose social origins (disciplinary context, career incentives, paradigmatic commitments) are invisible. Integration requires opening these boxes.

4 Five Structural Barriers to Integration

4.1 Theoretical Framework

Define scientific integration as the construction of a meta-framework $\mathcal{F}$ that:

1. Subsumes existing models $\{M_1, M_2, \ldots, M_n\}$ as special cases
2. Uses consistent notation and assumptions
3. Enables cumulative validation across domains
4. Generates novel predictions beyond component models

Integration is a public good: costly to produce, freely available once produced, and beneficial to all users. Standard public goods theory predicts underproduction.

4.2 A Simple Model

Let V_I denote the social value of integration and C_I its cost to the integrating scientist. Let p denote the probability of successful publication and career advancement.

Integration occurs if:

$$p \cdot V_{\text{private}} > C_I \tag{1}$$

where $V_{\text{private}} \ll V_I$ because most benefits of integration are external.

Under current academic incentives:

- p is low (peer review bias, disciplinary gatekeeping)
- V_{private} is low (integration papers don't fit career metrics)
- C_I is high (requires expertise across multiple fields)

The model predicts severe underproduction of integration.

4.3 Barrier 1: Academic Incentive Structures

$$\text{Academic Success} = f(\text{Publications}, \text{Citations}, \text{Grants}) \tag{2}$$

None of these metrics directly reward integration.

4.3.1 Smaldino & McElreath (2016): Natural Selection of Bad Science

Paul Smaldino and Richard McElreath developed an evolutionary model of science:

Model:

- Labs "reproduce" (train students) based on publication success
- Selection pressure favors methods that produce more publications
- Quality (replicability, integration) is not directly selected for

Key Finding:

> "As long as the incentives are in place that reward publication quantity... we expect that the rate of firing [researchers using poor methods] would have to be implausibly high to prevent the system from selecting for poor methods."

Implications:

1. Bad methods persist because they are *productive*

2. Good methods (like integration) are selected *against*
3. The system is in a stable equilibrium of fragmentation

4.3.2 Stephan (2012): How Economics Shapes Science

Paula Stephan's comprehensive analysis shows:

- Scientists respond rationally to incentives
- Incentives favor safe, incremental work
- High-risk integration is career-threatening
- The "winner-take-all" job market increases risk aversion

4.3.3 Edwards & Roy (2017): Perverse Incentives

Table 1: Perverse Incentives in Academia (Edwards & Roy, 2017)

Metric	**Perverse Effect**
Publication count	Salami-slicing, least publishable units
Journal impact factor	Chasing "hot" topics, avoiding integration
Citation count	Self-citation, citation cartels
Grant funding	Safe projects, avoiding risk
H-index	Gaming, quantity over quality

4.4 Barrier 2: Disciplinary Silos

4.4.1 The Structure of Silos

Disciplines develop specialized vocabularies, methods, and standards of evidence (Klein, 1990). These serve legitimate functions—enabling cumulative progress within paradigms—but create barriers to cross-disciplinary communication.

The same concept receives different names in different fields:

- "Habit" (psychology) = "Autopilot" (neuroscience) = "State-dependence" (economics)

- "Social proof" (psychology) = "Information cascade" (economics) = "Conformity" (sociology)
- "Ego depletion" (psychology) = "Self-control depletion" (economics) = "Decision fatigue" (management)

4.4.2 Jacobs (2013): In Defense of Disciplines

Jerry Jacobs argues that disciplines exist for good reasons:

- Shared methods enable cumulative progress
- Specialized training ensures expertise
- Peer review works within shared paradigms

But he also notes:

> "The very mechanisms that make disciplines productive also create barriers to the flow of ideas across fields."

4.4.3 Klein (1990): Interdisciplinarity

Julie Thompson Klein's comprehensive analysis identifies barriers:

- **Cognitive:** Different assumptions, methods, standards of evidence
- **Social:** Different communities, conferences, journals
- **Institutional:** Different departments, hiring criteria, tenure standards
- **Political:** Competition for resources, status hierarchies

4.4.4 Fortunato et al. (2018): Science of Science

Network analysis of 21 million papers shows:

- Citation networks cluster strongly by discipline
- Cross-disciplinary citations are rare and declining
- "Star" scientists bridge silos but are exceptions

Formally:

$$P(\text{cite across discipline}) \approx 0.1 \cdot P(\text{cite within discipline}) \tag{3}$$

4.5 Barrier 3: Peer Review Bias Against Integration

4.5.1 Uzzi et al. (2013): Atypical Combinations

Brian Uzzi and colleagues analyzed 17.9 million papers:

Finding 1: High-impact papers combine conventional and atypical knowledge.

Finding 2: Reviewers systematically penalize atypicality.

> "Papers that are highly novel are more likely to be rejected, take longer to get published, and are initially cited less—but eventually have higher impact if they survive."

Implication: Integration is high-risk, high-reward—but the risk is borne by the researcher, the reward accrues to society.

4.5.2 Wang et al. (2017): Bias Against Novelty

Jian Wang and colleagues document quantitative bias against novelty:

- Novel papers take 15% longer to publish
- Novel papers are 20% more likely to be desk-rejected
- Novel papers have higher variance in outcomes
- Effect sizes are larger for interdisciplinary work

4.5.3 The Reviewer's Dilemma

Typical reviewer reasoning for integration papers:

- "This is too broad—not rigorous enough for our discipline"
- "The author doesn't understand [my field] deeply enough"
- "This doesn't fit our journal's scope"
- "Where is the novel empirical contribution?"

Result: Integration papers are rejected by *all* disciplines because they fail to meet each discipline's specialized criteria.

4.6 Barrier 4: Funding Structures

Grant agencies reward well-defined, testable hypotheses. Integration is inherently risky: outcomes are uncertain, timelines are long, and success criteria are ambiguous. Risk-averse funding committees favor incremental projects.

Analysis of NIH and NSF funding patterns reveals:

- Success rates decline for interdisciplinary proposals
- Review panels lack expertise to evaluate integration
- Safe, incremental projects dominate portfolios
- Integration requires longer time horizons than typical 3-year grants

Junior researchers—who might have the energy for integration—face the strongest pressure to conform because the academic job market is winner-take-all.

4.7 Barrier 5: Territorial Defense

Paradigms are not merely intellectual structures but social institutions with defenders. Integration threatens established authority by suggesting that existing models are incomplete. Kuhn (1962) documented how paradigm defenders resist challenges.

Historical case studies document paradigm defense:

- **Behavioral economics** faced decades of resistance from rational choice theorists who controlled top journals
- **Plate tectonics** was rejected for 50 years despite geological evidence
- **Ulcer-bacteria theory** met fierce opposition from the acid-reduction paradigm
- **Prospect theory** took years to be accepted despite experimental evidence

The pattern is consistent: integration threatens established authority and faces resistance proportional to the stakes involved.

5 The Replication Crisis as Symptom

5.1 Ioannidis (2005): Why Most Research Findings Are False

John Ioannidis's landmark paper showed:

- For most study designs, most published findings are false
- Small sample sizes, small effect sizes, flexibility in analysis
- Publication bias: Positive results published, null results buried

Connection to Fragmentation: Each fragmented model is validated only within its silo. Without integration, there is no cumulative validation across frameworks.

5.2 Open Science Collaboration (2015)

The Reproducibility Project attempted to replicate 100 psychology studies:

- Only 36% replicated successfully
- Mean effect size dropped by 50%
- High-profile findings often failed

5.3 Camerer et al. (2016, 2018): Economics Replications

Similar findings in economics:

- 61% of experimental economics studies replicated (2016)
- 62% of social science papers in Nature/Science replicated (2018)

Implication: Fragmented models cannot be cumulatively validated. Integration would enable cross-model consistency checks that might catch replication failures earlier.

6 Scholarly Calls for Integration

6.1 Wilson (1998): Consilience

Edward O. Wilson's manifesto for unification:

> "The greatest enterprise of the mind has always been and always will be the attempted linkage of the sciences and humanities. The ongoing fragmentation of knowledge and resulting chaos in philosophy are not reflections of the real world but artifacts of scholarship."

Wilson argues that biology, psychology, and social science should converge—they study the same phenomenon (human behavior) at different levels.

6.2 Gintis (2009): The Bounds of Reason

Herbert Gintis explicitly calls for behavioral science unification:

> "The behavioral sciences should be unified... The current separation of economics, psychology, sociology, and anthropology is an artifact of disciplinary history, not a reflection of natural boundaries in the subject matter."

Gintis proposes:

- Shared mathematical framework (game theory + evolution)
- Common experimental methods
- Unified theory of human behavior

6.3 Bammer (2013): Integration and Implementation Sciences

Gabriele Bammer proposes a new field dedicated to integration:

- **Integration:** Combining knowledge from different disciplines
- **Implementation:** Applying integrated knowledge to real problems
- **I2S:** A discipline whose subject matter is integration itself

6.4 Other Calls

Table 2: Scholarly Calls for Integration

Author	**Work**	**Call**
Wilson (1998)	Consilience	Unity of knowledge
Gintis (2009)	Bounds of Reason	Unified behavioral science
Bammer (2013)	Disciplining Interdisciplinarity	Integration as discipline
Arthur (2015)	Complexity Economics	Complexity as unifier
D.S. Wilson (2019)	This View of Life	Evolution as meta-theory

7 The Framework versus Model Distinction

7.1 The Epistemological Core

Before examining how to address structural barriers, we must clarify a fundamental epistemological distinction that explains *why* economics fragmented.

Table 3: Framework versus Model

	Framework	**Model**
Purpose	Organizes thinking	Makes predictions
Claim	"Here is how to think about X"	"X equals Y"
Falsifiable?	No (meta-level)	Yes (testable)
On anomalies	Integrates them	Is falsified by them
Scope	Broad, organizing	Narrow, specific
Example	Newtonian mechanics as worldview	$F = ma$ as equation

7.2 The Map Analogy

A Framework is like a map style (topographic, political, road map).

A Model is like a specific route ("Take A1 to Bern, exit 24").

Map Style (Framework)	**Route (Model)**
Shows *what to look for*	Tells *what to do*
Multiple routes possible	One specific prediction
Can't be "wrong"—only useful or not	Can be wrong (road closed)
Integrates new roads easily	Fails if road doesn't exist

Key insight: When your route fails (traffic jam), you don't throw away the map. You use the map to find another route. But economics threw away the map when routes failed.

7.3 The Historical Confusion: Samuelson 1947

Paul Samuelson's *Foundations of Economic Analysis* (1947) was conceived as a **framework**—a mathematical language for expressing economic relationships. Samuelson himself noted the generality of the approach.

However, the profession **treated it as a model**:

- $C_{ij} = 0$ (separability) became a *testable assumption*
- Rational utility maximization became a *prediction*
- Equilibrium existence became an *empirical claim*

7.4 The Consequence: Fragmentation

When anomalies accumulated (Allais 1953, Kahneman-Tversky 1979, Güth 1982, Fehr-Schmidt 1999):

1. The "model" was falsified by experimental evidence
2. But no **framework** existed to integrate the anomalies
3. Each anomaly became its own literature (behavioral economics, identity economics, institutional economics...)
4. Result: **Fragmentation**

Classical economics was a **framework** conceived as a way of thinking. It was **treated as a model** with testable predictions. When predictions failed, **it could serve as neither**:

- Not as a model (falsified)
- Not as a framework (couldn't integrate anomalies)

Result: Economics lost its **gravitational center**.

8 The Chicago School: Documentation of the Framework-Model Confusion

8.1 The Chicago School Generations

The "Chicago School" was not monolithic but evolved through distinct generations, each reinforcing the treatment of separability ($C_{ij} = 0$) as empirical fact rather than simplifying assumption.

Table 4: Chicago School Generations and Key Figures

Generation	Key Figures	Contribution
Founders (1930s–50s)	Knight, Simons, Director, Viner	Established Chicago as distinct school
Core (1950s–70s)	Friedman, Stigler, Becker	"As if" methodology; economic imperialism
Rational Expectations (1970s–80s)	Lucas, Sargent, Prescott	Expectations always rational
Finance (1970s–90s)	Fama, Miller, Scholes	Efficient Markets Hypothesis
Law & Economics (1970s–)	Posner, Demsetz, Peltzman	Legal analysis as optimization

8.2 Friedman (1953): The "F-Twist"

Milton Friedman's essay "The Methodology of Positive Economics" is, according to Daniel Hausman, "the most influential work on economic methodology of [the twentieth] century."

The key passage:

> "Truly important and significant hypotheses will be found to have 'assumptions' that are wildly inaccurate descriptive representations of reality, and, in general, **the more significant the theory, the more unrealistic the assumptions**."

Paul Samuelson later termed this reasoning the "F-Twist"—the argument that shielded assumptions (including $C_{ij} = 0$) from empirical scrutiny.

8.3 Becker (1976): Economic Imperialism

Gary Becker's *The Economic Approach to Human Behavior* extended utility maximization to all domains:

> "All human behavior can be viewed as involving participants who **maximize their utility** from a stable set of preferences."

This transformed a framework assumption into an ontological claim about human nature.

8.4 Lucas (1976): Rational Expectations as Axiom

Robert Lucas's "Econometric Policy Evaluation: A Critique" established that expectations must be model-consistent:

$$\mathbb{E}[C|\mathcal{I}] = C^*(\Psi) \quad \text{always} \tag{4}$$

This ruled out systematic belief errors by *assumption*, not evidence.

8.5 The 2008 Crisis as Falsification

The financial crisis exposed the limitations of treating the framework as model:

> "Economists not only failed to anticipate the [financial] crisis but may have **contributed to it**—with risk and derivatives models that encouraged policy makers to see more stability than was actually present... [T]he crisis was met with incomprehension by most economists because of models that **downplay the possibility that economic actors may exhibit highly interactive behavior**." (Colander et al., 2009)

This is precisely the consequence of treating $C_{ij} = 0$ as fact rather than simplification: models could not capture the complementarities that drove systemic risk.

8.6 The Freshwater vs. Saltwater Divide

The methodological divide within economics has geographic correlates:

Table 5: Freshwater vs. Saltwater Economics

Freshwater (Chicago, Minnesota)	**Saltwater (MIT, Harvard, Berkeley)**
$C_{ij} = 0$ strictly	$C_{ij} \neq 0$ possible
Context exogenous, stable	Context partially endogenous
Markets always clear	Frictions accepted
Rational expectations axiom	Bounded rationality possible
Friedman, Lucas, Sargent, Prescott, Fama	Samuelson, Tobin, Solow, Stiglitz, Blanchard

9 The Case for Formalization

9.1 Why Mathematics Was Necessary

The preceding critique of the Chicago School's epistemological confusion must not be misread as a critique of mathematical formalization itself. On the contrary: formalization was **necessary** for economics to develop as a cumulative science.

9.2 Debreu's Defense of Rigor

Gérard Debreu, in his 1983 Nobel Lecture and 1991 Presidential Address to the American Economic Association, articulated the core benefits:

1. **Logical scrutiny:** "In its mathematical form, economic theory is open to an efficient scrutiny for logical errors."
2. **Cumulative development:** "The greater logical solidity of more recent analyses has contributed to the rapid contemporary construction of economic theory. It has enabled researchers to **build on the work of their predecessors**."
3. **Unambiguous communication:** Mathematics provides "an efficient language for unambiguous communication among economists."
4. **Secure foundations:** Axiomatization provides "secure bases from which exploration could start in new dimensions."

9.3 Concepts That Required Formalization

Many of economics' most important discoveries were *impossible* without mathematical formalization:

Table 6: Concepts Requiring Mathematical Formalization

Concept	**Discoverer(s)**	**Why Math Essential**
Adverse Selection	Akerlof (1970)	Equilibrium with asymmetric info
Signaling	Spence (1973)	Separating vs. pooling equilibria
Moral Hazard	Mirrlees (1971)	Principal-agent optimization
Nash Equilibrium	Nash (1950)	Simultaneous best responses
Mechanism Design	Hurwicz, Maskin, Myerson	"Reverse game theory"
Arrow Impossibility	Arrow (1951)	Logical proof of impossibility
Option Pricing	Black-Scholes-Merton	Stochastic differential equations
Auction Theory	Vickrey, Myerson	Revenue equivalence theorem
Contract Theory	Hart, Holmström	Optimal incentive structures
Matching Theory	Gale-Shapley, Roth	Stable matching algorithms

Without mathematical formalization, we would not know:

- Why high-quality sellers exit markets with asymmetric information
- How to design auctions that reveal true valuations
- When additional performance signals improve contracts
- That no voting system can satisfy all fairness criteria simultaneously

9.4 Rigor vs. "Mathiness"

Paul Romer (Nobel 2018) distinguished proper mathematical rigor from its abuse:

> "Mathiness... is a style that lets economists draw **invalid inferences** from the assumptions and structure of a model."

The distinction:

Rigor (Debreu-Style)	**Mathiness (Abuse)**
Mathematics makes assumptions explicit	Mathematics obscures assumptions
Conclusions follow logically	Invalid inferences from models
Limitations acknowledged	Limitations ignored
Framework $\neq$ Model	Framework = Model

9.5 The Real Critique

The critique of the Chicago School is **not**:

> "Economics should have used less mathematics."

The critique **is**:

> "Economics should have maintained the epistemological distinction between framework and model, treating $C_{ij} = 0$ as a *simplifying assumption* rather than an *empirical claim*."

The mathematics was correct. The epistemological confusion was the error.

10 Comparative Perspective: The Alternative to Formalization

To understand why formalization was necessary, consider what happened in social sciences that *did not* formalize.

10.1 Psychology: The Replication Crisis

Psychology, which relied less on mathematical formalization, experienced a devastating replication crisis:

- Only 36% of 100 studies replicated (Open Science Collaboration, 2015)
- "P-hacking" widespread (Simmons, Nelson, Simonsohn, 2011)
- Publication bias severe (Rosenthal, 1979: "file drawer problem")
- Major effects not robust: ego depletion, power posing, social priming

Without mathematical rigor: Decades of research on "ego depletion," "power posing," and "social priming" proved largely non-replicable. Verbal theories permitted ambiguity that formal models would have exposed.

10.2 Sociology: Paradigm Fragmentation

Sociology, explicitly rejecting mathematical formalization, exhibits:

- Multiple competing paradigms with no integration mechanism
- Persistent debates over "What did Weber *really* mean?"
- Limited policy influence compared to economics
- No cumulative theoretical development

As Randall Collins observed: "Sociology remains a discipline with many paradigms but no consensus."

10.3 Management Science: Fashion Cycles

Business and management research, with minimal mathematical formalization, exhibits fashion cycles rather than cumulative progress:

1980s: TQM → 1990s: Reengineering → 2000s: Six Sigma → 2010s: Agile → 2020s: Digital Transformation

Each generation "forgets" the previous. No cumulative knowledge base develops. Consultants sell "new" concepts that recycle old ideas.

10.4 The Comparative Verdict

Table 7: Formalized vs. Non-Formalized Social Sciences

Criterion	**Economics**	**Psychology**	**Sociology**	**Management**
Mathematical formalization	High	Low	Very low	Very low
Clear definitions	Yes	Partial	No	No
Cumulative development	Yes	Weak	No	No
Replicability	High	36%	Unknown	N/A
Policy influence	Very high	Moderate	Low	High (fashion)
Predictive power	Moderate	Weak	Very weak	None
Nobel Prize	Since 1969	No	No	No

The pattern is clear: Formalization correlates with cumulation and influence. Its absence correlates with fragmentation and marginalization.

11 Technological Preconditions

11.1 The Computational Constraint

The preceding analysis might suggest that the Chicago School made an avoidable epistemological error. This would be historically naive. The framework-model conflation was not merely intellectually convenient—it was **technologically inevitable** given the computational constraints of the era.

Table 8: Computational Technology and Model Complexity

Era	**Technology**	**Modeling Constraint**
1940s–1960s	Pencil, slide rule, early mainframes	Only analytically solvable models
1970s–1980s	Mainframes, early PCs	Simple numerical solutions possible
1990s–2000s	PCs, early internet	Monte Carlo, agent-based models emerge
2010s	Cloud computing, big data	Large-scale simulations feasible
2020s	AI/LLMs, massive compute	Context-dependent models operationalizable

11.2 Why $C_{ij} = 0$ Was Necessary

The separability assumption ($C_{ij} = 0$) was not chosen because economists believed it was true. It was chosen because it was **the only assumption that permitted analytical solutions**:

With $C_{ij} = 0$	**With $C_{ij} \neq 0$**
Utility separable across goods	Cross-effects everywhere
Individual optimization independent	Simultaneous optimization required
Equilibrium: solve n equations	Equilibrium: solve n^2 interactions
Analytical solutions possible	Numerical methods required
Pencil and paper sufficient	Computers required

11.3 Friedman's Defense Revisited

Friedman's "as if" methodology (1953) can be read not as epistemological confusion but as **pragmatic necessity**:

> *Given that we cannot compute models with realistic assumptions, we must use unrealistic assumptions that permit computation. The test is predictive success, not assumption realism.*

This was **defensible in 1953**. It became problematic only when the technological constraints lifted but the methodology persisted.

11.4 The Technological Milestones

Table 9: Technology Enabling Framework-Model Separation

Year	Technology	What It Enabled
1981	IBM PC	Econometrics democratized
1985	Spreadsheets	First simulations accessible
1991	World Wide Web	Data sharing, collaboration
2000s	Cloud computing	Massive Monte Carlo simulations
2007	Amazon MTurk	Cheap experimental subjects
2008	Smartphones	Real-time behavioral data
2017	Transformer architecture	Language models feasible
2022	ChatGPT/Claude	LLM Monte Carlo for context estimation

11.5 What AI Specifically Enables

The final technological piece—large language models—enables what was previously impossible: **operationalizing context-dependent models at scale**.

Table 10: AI-Enabled Capabilities for Integrative Frameworks

Component	**Before AI**	**With AI**
Context estimation	Manual surveys, expensive, slow	LLM Monte Carlo simulation
Context-specific models	Develop each separately	Generate automatically from framework
Heterogeneous preferences	2–3 discrete types	Continuous distributions
Cross-domain integration	Manual, inconsistent	Embedding-based consistency
Calibration	Months of fieldwork	Hours with LLM simulation

11.6 The Historical Verdict

Retrospectively: The framework-model conflation was epistemologically incorrect. Treating $C_{ij} = 0$ as empirical fact rather than simplifying assumption led to 50+ years of fragmentation.

Practically: The conflation was defensible given technological constraints. Without computers, models with $C_{ij} \neq 0$ were unsolvable. The Chicago methodology was a rational response to computational limitations.

Now: The technological constraints have lifted. Computers can solve models with $C_{ij} \neq 0$. AI can estimate context across situations. The framework-model distinction is now *operationally feasible*.

12 Integration Outside Academia: A Structural Prediction

12.1 The Prediction

If the barrier to integration is structural, integration should emerge from contexts with different structures. This prediction is consistent with observed patterns:

- Consulting firms integrate models for practical application
- Policy organizations need unified frameworks for decision-making
- Research institutes outside universities face different incentives

12.2 Why Integration Succeeds Outside Academia

1. **Different Incentives:** Consulting firms need working models, not publications
2. **Practice Pressure:** Transformation projects fail at 70%—integration is demanded
3. **Long Time Horizon:** 20+ years of iteration, not 3-year grant cycles
4. **Interdisciplinary Teams:** Economists + psychologists + practitioners
5. **No Territorial Stake:** No paradigm to defend

12.3 The Integration Proof

Claim: Integration is possible.

Evidence:

- Integrative frameworks exist and are used in practice
- Multiple disciplinary insights can be unified under common concepts
- Existing models become *special cases*, not competitors
- Framework + calibration data = predictive power

Conclusion: The barrier was institutional, not intellectual. The framework-model confusion prevented integration for 50+ years.

12.4 How Integrative Frameworks Address Each Barrier

Table 11: Addressing Structural Barriers

Barrier	**Response**
Academic incentives	Built outside academia; validated through practice
Disciplinary silos	Single conceptual framework spans all disciplines
Peer review bias	Validation through application, not publication
Funding structures	Funded by practice needs, not grants
Territorial defense	Shows models as *special cases*, preserving credit
Replication crisis	Integration enables cross-model consistency checks

13 The Metascience Movement

13.1 Key Institutions

- **Center for Open Science** (Brian Nosek): Pre-registration, open data
- **METRICS** (Stanford, Ioannidis): Meta-research on research
- **Research on Research Institute** (UK): Systemic analysis
- **Santa Fe Institute**: Complexity as integration principle

13.2 Key Reforms Proposed

1. **Pre-registration:** Commit to hypotheses before data collection
2. **Registered Reports:** Peer review before results known
3. **Open Data:** Share data for replication
4. **Multi-site Studies:** Replicate across contexts

5. **Meta-analysis:** Cumulative synthesis

13.3 What's Missing: Integration

The metascience movement focuses on *replication* and *transparency*—but not on *integration across paradigms*.

Integration fills this gap: It doesn't just replicate existing models, it **unifies** them under common organizing principles.

14 Economics as the Formalizable Social Science

14.1 The Central Thesis

Economics is the **only fully formalizable social science**—not because economists were smarter, but because they *pragmatically adapted* to technological constraints.

The adaptation produced epistemological confusion. But it also produced *the infrastructure* for a complete science of human behavior.

14.2 The Contrast with Other Disciplines

Table 12: Economics vs. Other Social Sciences: The Formalizability Thesis

Discipline	**Approach to Technology**	**Result**
Economics	Adapted methods to what was computable; formalized everything possible	Fully formalizable; cumulative; policy-relevant
Psychology	Rejected formalization as "reductionist"	Replication crisis; limited cumulation
Sociology	Explicitly rejected mathematical methods	Paradigm fragmentation; no integration
Political Science	Partial formalization (rational choice)	Split discipline; limited prediction
Anthropology	Rejected quantification entirely	Rich description; no generalization

14.3 The Infrastructure Argument

Seventy years of economic formalization created something irreplaceable: a **mathematical backbone** onto which insights from all social sciences can be integrated.

Integration can only emerge from the economic tradition **not** because:

- Economists are smarter (they are not)
- Economic insights are more important (all disciplines contribute)
- Economics "colonized" other fields (imperialism is not integration)

Integration can only emerge from economics **because**:

- Economics built the formal infrastructure over 70 years
- This infrastructure provides the backbone for integration
- Other social sciences lack this cumulative formal structure
- AI can operationalize this infrastructure—but cannot create it

The timing matters: The infrastructure had to exist *before* AI could operationalize it.

15 Discussion

15.1 The Core Paradox Revisited

The central paradox of behavioral science fragmentation can be stated simply:

> *Everyone agrees integration is needed. No one is rewarded for doing it.*

This is not a coordination failure that can be solved by agreement. It is a structural problem requiring structural solutions.

15.2 Limitations

This analysis has limitations:

- Evidence is largely observational; causal identification is difficult
- Quantitative estimates vary across studies and contexts
- Counter-examples exist (successful interdisciplinary fields)
- The analysis focuses on barriers, not detailed solutions

15.3 Future Research

Promising directions include:

- Experimental studies of peer review for integrative work
- Analysis of successful integration cases
- Design of institutional mechanisms that reward integration
- Evaluation of alternative metrics for scientific contribution
- Longitudinal studies of framework adoption in practice

15.4 Institutional Innovation

Overcoming structural barriers requires institutional innovation:

Alternative incentive structures: Organizations that reward integration directly, not through publication metrics.

Boundary-spanning roles: Scientists who specialize in translation across disciplines, valued for integration rather than within-discipline contribution.

Different funding mechanisms: Support for long-term, high-risk integration projects with appropriate evaluation criteria.

New publication venues: Journals that value integration, with review processes appropriate for interdisciplinary work.

16 Conclusion

The fragmentation of behavioral science into 40+ competing models is not an intellectual failure but a structural outcome. Five mechanisms—academic incentives, disciplinary silos, peer review bias, funding structures, and territorial defense—systematically prevent integration despite universal recognition of its value.

A central epistemological error—treating Samuelson's framework as a falsifiable model—contributed to fragmentation when behavioral anomalies accumulated without an integrative structure to absorb them. This error was technologically defensible when only analytically solvable models were computationally feasible, but became problematic as computational constraints lifted.

The barrier to integration is institutional, not intellectual. Integration is technically possible—the existence of integrative frameworks demonstrates this. But academic institutions as currently structured will not produce it.

This analysis suggests that integration will continue to emerge from outside traditional academic structures: consulting firms, policy organizations, and research institutes with different incentive systems. The Evidence-Based Framework for Economic and Social Behavior exemplifies this pattern: built outside academia, validated through practice,

and designed to unify rather than compete.

For behavioral science to achieve cumulative integration, institutional innovation is required. The science of science has documented the problem; designing solutions remains for future work. The technological preconditions for integration now exist—what remains is the institutional will to pursue it.

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A Data Sources

This paper synthesizes findings from the following primary data sources:

- Web of Science citation database (Fortunato et al., 2018)
- NIH and NSF grant databases (Stephan, 2012)
- Peer review outcome data (Wang et al., 2017)
- Agent-based simulations (Smaldino & McElreath, 2016)
- Open Science Collaboration replication data (2015)
- Economics replication data (Camerer et al., 2016, 2018)

B Historical Timeline

Table 13: Key Events in the Framework-Model Confusion

Year	Event
1947	Samuelson publishes *Foundations of Economic Analysis*
1953	Friedman publishes "Methodology of Positive Economics"
1959	Arrow-Debreu prove general equilibrium existence
1962	Kuhn publishes *Structure of Scientific Revolutions*
1970	Akerlof publishes "Market for Lemons"
1976	Lucas Critique published
1979	Kahneman-Tversky publish Prospect Theory
1999	Fehr-Schmidt publish inequality aversion model
2005	Ioannidis publishes "Why Most Research Findings Are False"
2008	Financial crisis exposes model limitations
2015	Open Science Collaboration reports 36% replication rate
2016	Smaldino & McElreath publish "Natural Selection of Bad Science"
2020s	AI enables context-dependent modeling at scale

C Relationship to EBF

This working paper was generated from Appendix LIT-META of the Evidence-Based Framework documentation. The appendix contains additional detail on:

- The 8Ψ Model of Scientific Development
- Beatrix operationalization of the framework-model distinction
- Extended historical case studies
- Quantitative citation network analysis
- EBF-specific responses to metascience critique

For the complete analysis, see Appendix LIT-META in the EBF documentation.